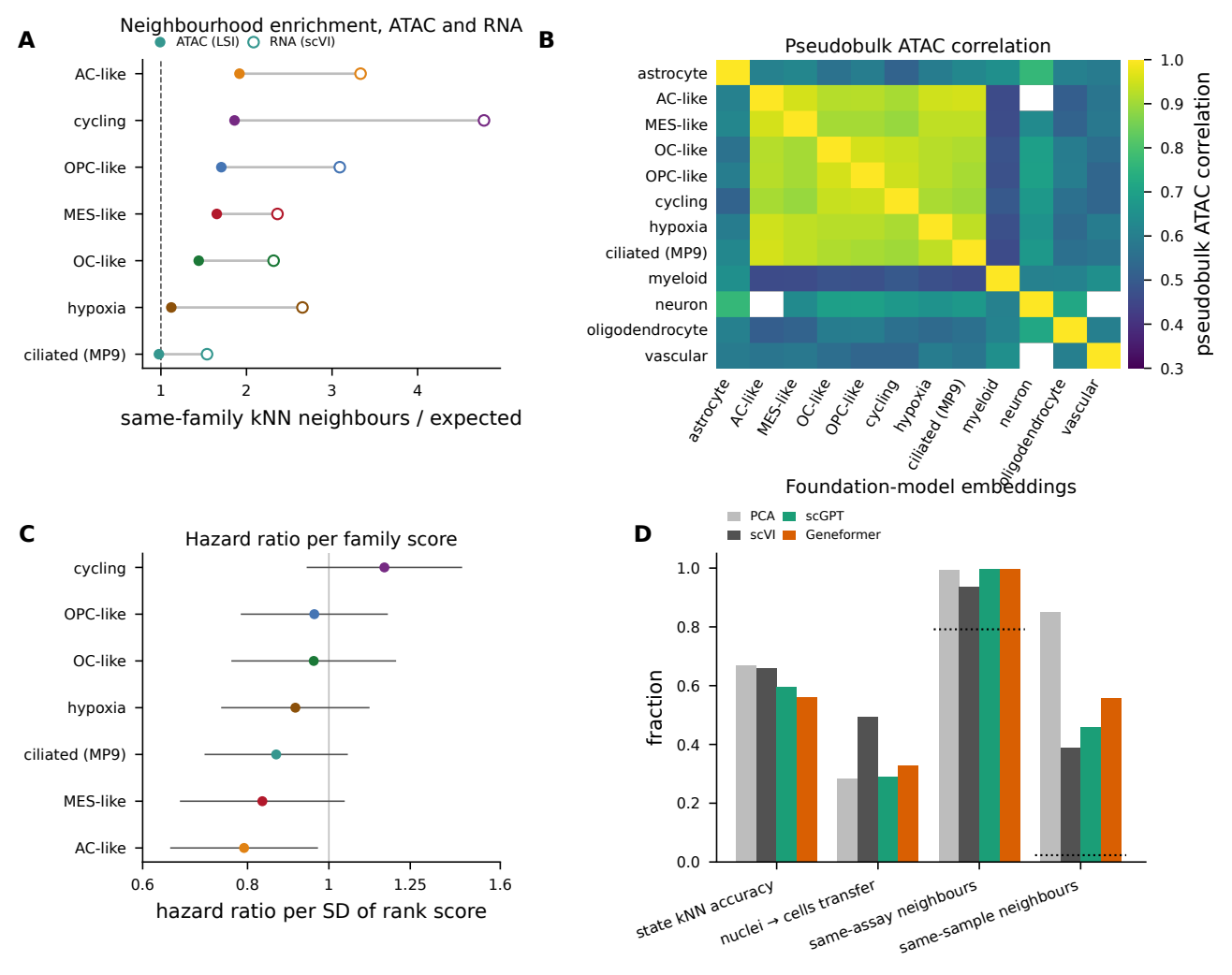
A
Neighbourhood enrichment, ATAC and RNA
ATAC (LSI)
RNA (scVI)
AC-like
cycling
OPC-like
MES-like
OC-like
hypoxia
ciliated (MP9)
1
2
3
4
same-family kNN neighbours / expected
B
Pseudobulk ATAC correlation
astrocyte
AC-like
MES-like
OC-like
OPC-like
cycling
hypoxia
ciliated (MP9)
myeloid
neuron
oligodendrocyte
vascular
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
pseudobulk ATAC correlation
C
Hazard ratio per family score
cycling
OPC-like
OC-like
hypoxia
ciliated (MP9)
MES-like
AC-like
0.6
0.8
1
1.25
1.6
hazard ratio per SD of rank score
D
Foundation-model embeddings
PCA
scVI
scGPT
Geneformer
1.0
0.8
0.6
0.4
0.2
0.0
fraction
state kNN accuracy
nuclei → cells transfer
same-assay neighbours
same-sample neighbours